

Bommidi Sai Pranith

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Career Objective

ECE undergraduate with strong interest in wireless communications, signal processing, and machine learning. Seeking a research internship to apply concepts such as modulation, beamforming, and channel modeling in advanced communication systems.

Education

KLH University

Sept 2023 – May 2027

B.Tech in Electronics and Communication Engineering

CGPA: 7.72

SR Junior College

2021 – 2023

Intermediate (MPC)

Percentage: 96%

Mary Mediatrix High School

2021

SSC

CGPA: 10.0

Relevant Coursework

Digital Communication, Signals and Systems, Analog Communication, Probability and Random Processes, Electromagnetic Theory

Technical Skills

Core Concepts: Digital Communication (Modulation, BER, Noise Analysis), Signal Processing (Fourier Transform, Filtering, Sampling), Wireless Communication (Fading, Path Loss, SNR, EIRP), 5G Concepts (Beamforming, mmWave, MIMO)

Programming: Python (NumPy, basic simulations), C (fundamentals)

Tools: Ansys STK, MATLAB (basic), Git

Projects

Hybrid Beamforming in mmWave Systems

- Simulated hybrid beamforming for massive MIMO systems using Python.
- Compared analog, digital, and hybrid beamforming techniques.
- Analyzed improvements in signal strength and interference reduction.

Satellite Communication Link Budget Analysis

- Studied parameters such as EIRP, G/T ratio, and path loss.
- Evaluated impact of atmospheric losses on signal quality.
- Analyzed noise temperature and SNR in satellite communication systems.

Digital Communication System Simulation

- Simulated BPSK and QPSK modulation schemes.
- Evaluated Bit Error Rate (BER) under AWGN conditions.
- Studied effects of noise and channel impairments.

Certifications

- STK Certification (Level 1) – Ansys
- Introduction to Networking – NVIDIA Academy
- Introduction to Cybersecurity – Cisco Networking Academy
- Ansible Essentials for Network Engineers – NVIDIA Academy

Technical Interests

Wireless Communication Systems, Signal Processing, 5G and mmWave Technologies, Satellite Communication, Machine Learning for Communication Systems

Additional Information

- Strong interest in research-oriented problem solving
- Ability to work independently and in collaborative environments
- Actively learning advanced wireless communication concepts